

Association between electroconvulsive therapy and time to readmission after a manic episode

Katarzyna Popiolek¹  | Tor Arnison¹ | Susanne Bejerot¹ | Katja Fall¹ | Mikael Landén^{2,3} | Axel Nordenskjöld¹ 

¹University Health Care Research Centre, Faculty of Medicine and Health, Örebro University, Örebro, Sweden

²Institute of Neuroscience and Physiology, The Sahlgrenska Academy, Gothenburg University, Gothenburg, Sweden

³Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, Stockholm, Sweden

Correspondence

Katarzyna Popiolek, Department of Psychiatry, Örebro University Hospital, 701 85 Örebro, Sweden.
Email: katarzyna.popiolek@regionorebrolan.se

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Abstract

Objective: The majority of patients hospitalized for treatment of a manic episode are readmitted within 2 years despite maintenance treatment. Electroconvulsive therapy (ECT) has been associated with lower rehospitalization rates in some psychiatric conditions, but its association with readmission after a manic episode has not been investigated. Therefore, the aim of this study was to determine whether the time to readmission in patients with mania treated with ECT was longer than in patients not treated with ECT and whether there were subgroups of patients that benefited more.

Methods: This was a nationwide register-based, observational study. All patients diagnosed with bipolar disorder, manic episode, admitted to any hospital in Sweden between 2012 and 2021 were included. Patients contributed data to the study for every admission. All admissions were followed up until psychiatric readmission, death, or the end of the study (December 31, 2021). Association between ECT and time to readmission was analyzed. A paired samples model was performed for 377 patients with at least two admissions for mania, treated with ECT at one admission and without ECT at the other admission. Times to readmission were analyzed.

Results: A total of 12,337 admissions were included; mean (SD) age 47.7 (17.2), 5443 (44.1%) men. Readmission rate within 1 year was 54.6%. ECT was administered in 902 (7.3%) admissions. Within 30 days after admission, 182 out of 894 (20.4%) patients treated with ECT versus 2105 out of 11,305 (18.6%) patients treated without ECT were readmitted. There was no association between ECT and time to readmission (aHR 1.00, 95% CI 0.86–1.16, $p = 0.992$) in the model with all admissions. The paired samples model included 754 admissions (377 patients), mean (SD) age during admission without ECT was 45.6 (16.5), and with ECT 46.6 (16.4), 147 (39.0%) were men. In that model,

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readmission rate within 30 days for treatment with ECT was 19.0%, and for treatments without ECT, 24.1% (aHR 0.75, 95% CI 0.55–1.02, $p = 0.067$).

Conclusion: Readmission rates after inpatient treatment of mania were high. ECT was not significantly associated with longer time to readmission, but there was a trend toward a protective effect of ECT when admissions with and without ECT were compared within the same patients.

KEYWORDS

bipolar disorder, electroconvulsive therapy, mania, readmission, rehospitalization

1 | INTRODUCTION

Bipolar disorder is a chronic neuropsychiatric disorder characterized by episodes of depression and mania. The first line of treatment for manic episodes is psychopharmacotherapy: mood stabilizers, atypical antipsychotics, or combinations of these.^{1–4} Electroconvulsive therapy (ECT) is highly effective in mania with response rates reported at 85%.^{5,6} Nevertheless, ECT is recommended as a second- or third-line treatment for mania, mostly due to concerns for adverse effects, especially cognitive impairment, limited availability, and stigma.^{6,7} According to several studies, 50% of patients with mania need rehospitalization within 2 years despite maintenance treatment with pharmacotherapy.^{8,9} Some clinical factors have been associated with higher relapse rates: younger age at the first episode of mania,¹⁰ psychotic symptoms at the index admission,¹¹ higher number of psychiatric admissions before index admission,¹¹ and residual symptoms at recovery.^{9,12}

Some studies suggest that ECT in patients with mood disorders may have a positive impact on the risk of readmission.^{7,13–15} However, no study has investigated the association between ECT and readmission rates in patients with a recent manic episode.

The Swedish National Quality Register for ECT and other Swedish national registers offer the opportunity to study a unique large population of patients who have received ECT for mania. Many of the clinical factors that have been associated with relapse in previous studies can be estimated by variables in the Swedish registers. Therefore, it is possible to study the association between ECT and time to readmission following hospitalization for mania.

1.1 | Aim

This register-based, nationwide cohort study has two major aims: (i) to investigate time to readmission after a manic episode in two groups of patients: those treated with psychopharmacotherapy without ECT and those

Significant outcomes

- In this nationwide, register-based cohort study that included over 12,000 admissions for treatment of a manic episode, 55% of patients were readmitted within 1 year.
- Patients treated with ECT were more severely ill compared to those who were not treated with ECT, still there was no difference between time to readmission.
- In a paired samples model, there was a trend toward a longer time to readmission when individuals received ECT.

Limitations

- There was no randomization, thus residual indication bias might have affected the results of this study despite the methods used to balance admissions with and without ECT.
- Rating scales to assess severity of manic symptoms at discharge were not obtained.
- Information on maintenance ECT treatment after discharge was lacking.

treated with ECT; (ii) to investigate interactions between clinical and sociodemographic factors and the association between ECT and readmission after a manic episode. We hypothesize that patients in the ECT group would exhibit longer time to readmission and that this association would be strongest among the most severely ill patients.

2 | MATERIALS AND METHODS

2.1 | Design

This was a nationwide, register-based, cohort study. Data from several national registers were compiled using the

personal identity number. This data was used to identify the case group of patients treated with ECT for a manic episode, as well as the control group of all patients treated without ECT for a manic episode during inpatient care. Additionally, a self-control study design was applied in the paired sample models. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting for observational studies.¹⁶

2.2 | Ethics statement

The study was approved by the ethical review authority (2021–03815). All data were pseudonymized, and individuals were not identifiable at any time; thus, participants were not informed of the study and were not asked to provide consent.

2.3 | Participants

In this study, all patients admitted to any hospital in Sweden between January 1, 2012 and December 31, 2021 with the diagnosis of bipolar disorder, manic episode, were included. Patients contributed data to the study for every admission. Patients were diagnosed according to the International Classification of Diseases, Tenth Revision (ICD-10).¹⁷

2.4 | Outcomes

All patients were followed up until psychiatric readmission, or death, or the end of the study (December 31, 2021). The outcome was defined as readmission due to any psychiatric diagnosis after discharge. Readmission rates within different periods (1, 3, 6, and 12 months) were described for patients treated with and without ECT. For these analyses, cases that died or reached the end of the study before readmission within the analyzed period were excluded. The association between ECT and time to outcome within 30 days was analyzed for the whole study population and for subgroups stratified by sex, age, education level, cohabiting status, number of admissions, psychiatric comorbidities substance use disorder, anxiety disorders, obsessive-compulsive disorder (OCD), attention deficit hyperactivity disorder (ADHD), autism spectrum disorder (ASD), and maintenance psychopharmacological treatments, namely lithium, lamotrigine, valproate, antidepressants, oral antipsychotics (OAs), long-lasting antipsychotics (LAAs), and benzodiazepines to investigate whether ECT had beneficial effect

on time to readmission for the whole population and to identify subgroups of patients that benefited more. The same approach was used in the paired samples model. Suicide during the follow-up period was investigated for all admissions. As patients were followed up until readmission, death, or the end of the study, this analysis did not include suicides after readmission.

2.5 | Paired samples model

The retrospective study design is open to indication bias. We hypothesized that patients treated with ECT were more severely ill, hence the risk that a potential protective effect of ECT might be underestimated. We therefore also conducted a paired samples analysis. In this model, two hospitalizations for the treatment of manic episodes within the same patient were selected. The patient was treated with ECT during one of these admissions, but received other antimanic treatment than ECT during the other admission. Thus, patients in this model were controls for themselves. Furthermore, admissions with and without ECT were matched by the number of previous admissions, as this was strongly associated with time to readmission in this cohort (Appendix C) and in other studies.¹¹ We included two consecutive admissions in this model where ECT was allowed to be the first or the second. Outcomes for both admissions were then compared. This model reduced indication bias since patients with a more severe illness course were not compared with patients who had a milder illness course. However, this model did not account for the severity of the episode.

2.6 | Propensity score matching analysis

Additionally, a propensity score matching analysis was performed, using an optimal matching algorithm with 0.1 caliper width, to balance admissions with ECT and without ECT. For every admission with ECT (exposure group), two admissions without ECT (control group) were matched based on propensity score calculated from several variables: sex, age, education, cohabiting status, number of admissions before index admission, psychiatric comorbidities, and maintenance psychopharmacological treatments. The association between ECT and time to readmission within 30 days was analyzed for this population.

2.7 | Data sources and variables

The study was based on data from several national Swedish registers: (i) the Swedish National Quality Register for

ECT (Q-ECT), which provides information on ECT settings, and assessments before, during, and after treatment,¹⁸ (ii) the Swedish National Patient Register, which provides information on all hospital admissions and doctors' appointments in specialized outpatient care, diagnoses and procedures, age and sex,¹⁹ (iii) the Longitudinal Integration Database for Health Insurance and Labor Market Studies (LISA), which provides detailed information on socioeconomic status,²⁰ and (iv) the Swedish Prescribed Drug Register, which provides information on all filled drug prescriptions in Swedish pharmacies.²¹

Linkage of information across the registers by personal identity number gave a unique nationwide database.

Data about education came from LISA with information up to 2019. In most cases, the highest education level documented at the index admission was used. For admissions after 2019, the highest education level in 2019 was used. In 476 cases, information about the highest education level at admission was missing. In 52 cases, information about education was available the following year and was used instead, leaving 424 (3.1%) cases with missing information that were imputed to the largest group (high school education).

Data about cohabiting came from LISA with information up to 2019 and referred to living with a partner regardless of marital status. For admissions after 2019, information about cohabiting from 2019 was used. In 264 (1.9%) cases, information about cohabiting during the admission year was missing. In 30 cases that information was available the following year and was used instead, leaving 234 (1.7%) of cases with missing information, which were imputed into the largest group (not cohabiting).

Information about the presence of psychotic symptoms came from the Swedish National Patient Register and was based on the diagnosis at discharge. Patients were diagnosed according to the ICD-10 with the codes F301 and F311 for manic episodes without psychotic symptoms or F302 and F312 for manic episodes with psychotic symptoms.

Information about pharmacological maintenance treatment came from the Swedish Prescribed Drug Register. Patients were classified as treated with a drug when they collected this drug at least once during admission or within 90 days after admission. Antipsychotics were divided into oral antipsychotics (OA) and long-acting injectable formulations (LAAs). The proportion of maintenance treatments with different OAs and LAAs is shown in Appendix A and Appendix B.

Information about comorbidities came from the Swedish National Patient Register.

2.8 | ECT devices

ECT was administered using bidirectional constant-current, brief-pulse Mecta (Mecta Corp., Tualatin, OR) or Thymatron System IV (Somatics LLC, Lake Bluff, IL) devices.

2.9 | Statistical analyses

Differences between admissions with and without ECT were described using standardized mean difference (SMD). SMD of >0.2 was used as a cut-off for indicating differences in key variables between the groups. Univariate and multivariable Cox regression analyses were used to investigate the association between clinical and socio-demographical factors and time to readmission during the 30 days after discharge. Cases were censored on day 31 after discharge, in case of death, or on December 31, 2021. Times to readmission after admissions with ECT and without ECT were then compared in subgroups stratified according to several variables (sex, age, education level, cohabiting status, number of admissions, psychiatric comorbidities, and maintenance psychopharmacological treatments) to identify subgroups of patients that have differential responses to ECT treatment. In this model, a multivariable Cox regression analysis was used with ECT treatment as an independent variable and all variables used in the main regression analysis as potential confounders. In separate models, interaction terms between the investigated variable and ECT treatment were added to determine the significance of the association. Readmissions during the first year after admissions with and without ECT in the paired model sample were illustrated using a Kaplan–Meier curve. The statistical significance of the difference between the distributions was tested with the Log Rank (Mantel-Cox) test. Additionally, the propensity score matching was performed, and the statistical significance of the differences between the time to readmission in ECT treated patients, and the control group of patients treated without ECT, was analyzed.

3 | RESULTS

3.1 | Participants

A total of 6013 individuals contributed to 12,337 admissions for mania between 2012 and 2021. ECT was administered in 902 (7.3%) admissions. The socio-demographic characteristics of the study population are presented in

TABLE 1 Socio-demographic and clinical characteristics of the cohort.

		All cases <i>N</i> = 12,337 <i>n</i> (%)	Without ECT <i>N</i> = 11,435 (%)	ECT treated <i>N</i> = 902 (%)	SMD*
Sex	Male	5443 (44.1)	5083 (44.5)	360 (39.9)	−0.068
Age (mean, SD)		47.7 (17.2)	47.8 (17.2)	46.4 (16.6)	−0.071
Education	Less than high school	2348 (19.0)	2172 (19.0)	176 (19.5)	-
	High school	5997 (48.6)	5547 (48.5)	450 (49.9)	0.014
	College <2 years	885 (7.2)	829 (7.2)	56 (6.2)	−0.054
	College ≥ 2 years	3107 (25.2)	2887 (25.2)	220 (24.4)	−0.018
Cohabiting	Yes	5352 (43.4)	4948 (43.3)	404 (44.8)	0.016
	No	6985 (56.6)	6487 (56.7)	498 (55.2)	-
Presence of psychotic symptoms	Yes	7221 (58.5)	6638 (58.0)	583 (64.6)	0.114
	No	5116 (41.5)	4797 (42.0)	319 (35.4)	-
Comorbidity	Anxiety disorder	2903 (23.5)	2689 (23.5)	214 (23.7)	0.016
	OCD	150 (1.2)	134 (1.2)	16 (1.8)	0.043
	Personality disorder	1362 (11.0)	1265 (11.1)	97 (10.8)	< −0.001
	ADHD	776 (6.3)	722 (6.3)	54 (6.0)	< −0.001
	Autism	298 (2.4)	273 (2.4)	25 (2.8)	0.026
	Substance abuse disorder	2152 (17.4)	1990 (17.4)	162 (18.0)	0.024
Length of index admission in days (median, IQR)		20 (8–32)	18 (7–31)	31 (22–50)	0.254*
Number of admissions before index admission (median, IQR)		8 (3–19)	8.0 (3–18)	11 (5–25)	0.273*
Maintenance therapy post-discharge	Antidepressants	2024 (16.4)	1878 (16.4)	146 (16.2)	−0.110
	Lithium	6404 (51.9)	5844 (51.1)	560 (61.1)	0.414*
	Benzodiazepines	3009 (24.4)	2774 (24.3)	235 (26.1)	0.002
	Antipsychotics oral	8808 (71.4)	8135 (71.1)	673 (74.6)	0.060
	Antipsychotics LAAs	1239 (10.0)	1112 (9.7)	127 (14.1)	0.118
	Lamotrigine	1292 (10.5)	1204 (10.5)	88 (9.8)	0.070
	Valproate	3716 (30.1)	3438 (30.1)	278 (30.8)	0.065
	Any antipsychotic	9343 (75.7)	8622 (75.4)	721 (79.9)	0.089
	Lithium +antipsychotic	5165 (41.9)	4707 (41.2)	458 (50.8)	0.157
	Valproate +antipsychotic	3105 (25.2)	2872 (25.1)	233 (25.8)	0.013

Abbreviations: ADHD, attention deficit hyperactivity disorder; ECT, electroconvulsive therapy; IQR, interquartile range; LAA, long-acting antipsychotics; OCD, obsessive compulsive disorder; SD, standard deviation; SMD, standardized mean difference;

*Values of >0.2 indicate differences between the groups.

Table 1. Patients treated with ECT had longer index admissions, had more admissions due to psychiatric disorders before the index admission, and were more often-prescribed lithium after discharge than patients not treated with ECT.

3.2 | ECT settings

The median number of ECT treatments was 6 (IQR 5–9, min 1, max 42). Most patients received three ECT sessions per week (the customary frequency in Sweden).

Decisions about electrode placement, stimulus dose, and number of sessions were made by clinicians based on patients age, sex, concomitant pharmacological treatment, and severity of symptoms. Electrical stimulus dose titration is uncommon in Sweden; instead, the quality of seizures and the clinical response are repeatedly evaluated to adjust ECT settings and anesthetics. The electrode placement at the first treatment session was unilateral in 52.2% of cases and bilateral in 47.8%. The last treatment session was unilateral in 63.9% of cases and bilateral in 36.1%. The median pulse width was 0.5 (IQR 0.5–0.5) at the first and 0.5 (IQR 0.5–0.6) at the last treatment session. The mean stimulation dose was 386 mC (SD, 171 mC) at the first and 456 mC (SD, 201 mC) at the last treatment session. The mean seizure duration according to EEG was 45 s (SD, 23 s) at the first and 38 s (SD, 17 s) at the last treatment session.

The median [IQR] time from the first admission day to the first day with ECT was 7 days [3–16] among admissions with ECT. Out of 902 admissions with ECT, 772 (85.6%) were assessed with the Clinical Global Impression-Improvement scale (CGI-I) within 1 week after the last ECT treatment in the series. CGI-I is a 7-point scale that is used by clinicians to evaluate the change due to the treatment. The vast majority of patients, 85.9% (663 individuals), were assessed as very much improved (CGI-I 1) or much improved (CGI-I 2) after ECT.

3.3 | Outcome

The mean time of follow-up, (time to readmission, death, or the end of the study) was 620.9 days. During the follow-up, 71.0% of the study population (8762 admissions) were readmitted. The median [IQR] time to readmission was 145 [29–428] days. The percentage of these that were readmitted within 1, 3, 6, and 12 months after discharge from the hospital is presented in Table 2. Within 1 month, 20.4% of patients treated with ECT and 18.6% of patients without ECT were readmitted. The most common diagnosis during readmission was mania (53.6%); a depressive episode was diagnosed in 10.3% of readmissions, and a mixed episode in 2.8% of cases; in 18.4% of readmissions, an affective episode was diagnosed without specification of polarity. All causes of readmissions are presented in Figure 1.

Suicide was rather uncommon after hospitalization for mania and occurred after 26 out of 12,337 admissions. In total, 65.4% of the cases (17 patients) were men. The median [IQR] age was 41 years [31–53]. Only one out of 26 patients who committed suicide after discharge from the hospital was treated with ECT during the index admission. In this case, suicide occurred 1493 days after discharge.

There was no association between ECT and time to readmission (aHR 1.00, 95% CI 0.86–1.16, $p = 0.992$) in the model with all admissions for mania. Associations between

TABLE 2 Readmission within different periods.

	All hospitalizations	ECT treated	Without ECT
Readmission within 1 month	18.7% (2287 of 12,199)	20.4% (182 of 894)	18.6% (2105 of 11,305)
Readmission within 3 months	30.3% (3642 of 12,009)	34.5% (300 of 870)	30.0% (3342 of 11,139)
Readmission within 6 months	40.9% (4816 of 11,765)	47.2% (401 of 850)	40.4% (4415 of 10,914)
Readmission within 12 months	54.6% (6228 of 11,411)	60.5% (493 of 815)	54.1% (5735 of 10,596)

Abbreviation: ECT, electroconvulsive therapy.

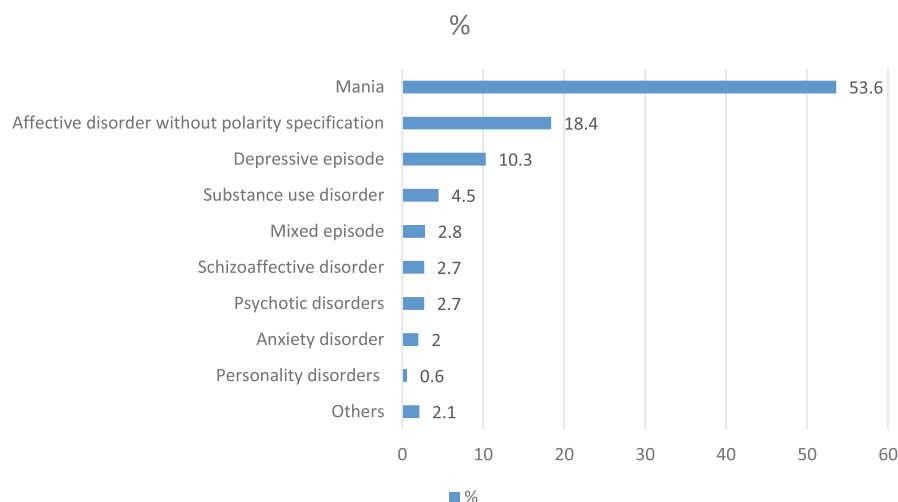


FIGURE 1 Diagnosis during readmission.

TABLE 3 Results of multivariable Cox regression analysis of association between ECT treatment and time to readmission within 30 days for different subpopulations.

	aHR	P-value	Interaction
Sex			
Male	1.14 (0.92–1.41)	0.236	Reference
Female	0.90 (0.73–1.11)	0.305	0.153
Age			
Age 12–45	1.12 (0.92–1.38)	0.266	Reference
Age 46–96	0.92 (0.74–1.15)	0.464	0.158
Education			
Less than high school	0.93 (0.66–1.30)	0.659	Reference
High school	1.05 (0.86–1.29)	0.645	0.632
Some college <3 yrs	1.28 (0.72–2.26)	0.402	0.386
College ≥ 3 yrs	0.90 (0.63–1.27)	0.536	0.412
Cohabiting			
No	0.90 (0.73–1.10)	0.285	Reference
Yes	1.13 (0.90–1.41)	0.303	0.079
Psychosis			
No	1.04 (0.81–1.33)	0.764	Reference
Yes	0.99 (0.82–1.20)	0.938	0.606
Substance use disorder			
No	1.07 (0.90–1.26)	0.452	Reference
Yes	0.81 (0.58–1.12)	0.199	0.175
Anxiety			
No	1.01 (0.84–1.20)	0.939	Reference
Yes	1.01 (0.76–1.334)	0.949	0.773
Personality disorder			
No	0.99 (0.84–1.16)	0.865	Reference
Yes	1.02 (0.71–1.46)	0.929	0.307
OCD			
No	1.00 (0.86–1.16)	0.994	Reference
Yes	1.49 (0.34–6.54)	0.599	0.680
ADHD			
No	1.04 (0.89–1.21)	0.628	Reference
Yes	0.66 (0.36–1.21)	0.180	0.187
Autism			
No	1.00 (0.86–1.16)	0.962	Reference
Yes	1.35 (0.55–3.29)	0.513	0.678
Number admissions			
0	0.72 (0.26–1.98)	0.528	Reference
1–2	1.02 (0.58–1.79)	0.960	0.435
3–5	0.91 (0.55–1.49)	0.699	0.882
6–10	1.33 (0.95–1.85)	0.097	0.602
11–20	0.79 (0.56–1.13)	0.192	0.277
>20	1.06 (0.84–1.34)	0.622	0.162

TABLE 3 (Continued)

	aHR	P-value	Interaction
Lithium			
No	0.85 (0.66–1.09)	0.199	reference
yes	1.13 (0.93–1.36)	0.218	0.042
Valproate			
No	0.97 (0.81–1.17)	0.778	reference
Yes	1.07 (0.83–1.38)	0.611	0.438
Lamotrigine			
No	0.99 (0.84–1.16)	0.873	Reference
Yes	1.14 (0.72–1.81)	0.566	0.589
Benzodiazepines			
No	1.03 (0.86–1.23)	0.779	Reference
Yes	0.96 (0.73–1.27)	0.780	0.661
LAAs			
No	1.03 (0.86–1.23)	0.772	Reference
Yes	0.96 (0.72–1.23)	0.760	0.822
Oral antipsychotics			
No	1.05 (0.78–1.40)	0.756	Reference
Yes	0.99 (0.83–1.18)	0.887	0.999
Antidepressants			
No	0.99 (0.84–1.17)	0.886	Reference
Yes	1.11 (0.77–1.61)	0.578	0.977

Abbreviations: ADHD, attention deficit hyperactivity disorder; aHR, adjusted hazard ratio; LAAs, long-acting antipsychotics; OCD, obsessive compulsive disorder; SD, standard deviation.

TABLE 4 Characteristics of the population of patients that were treated with and without ECT at two separate admissions.

	Admission without ECT	Admission with ECT	SMD
Age	45.6 (SD 16.5)	46.6 (SD 16.4)	0.047
Sex (male)	147 (39.0%)		
Number of admissions before index admission	Median 10	Median 11	0.054
Length of index admission (days)	Median 19	Median 33.0	0.431
Psychotic symptoms	219 (58.1%)	243 (64.5%)	0.108
Post-discharge pharmacotherapy			
Lithium	237 (62.9%)	237 (62.9%)	0
Valproate	107 (28.4%)	116 (30.8%)	0.043
Lamotrigine	42 (11.1%)	48 (12.7%)	0.040
Benzodiazepines	92 (24.4%)	92 (24.4%)	0
LAAs	46 (12.2%)	50 (13.3%)	0.027
Oral antipsychotics	279 (74.0%)	276 (73.2%)	−0.015

Abbreviations: ECT, electroconvulsive therapy; LAAs, long-acting antipsychotics; SD, standard deviation; SMD, standardized mean difference.

demographical, clinical, and social factors and time to rehospitalization or suicide within 30 days are presented in Appendix C. Subgroup analyses of the association between

ECT and readmission are presented in Table 3. None of the analyzed factors were significantly associated with differential outcomes after ECT treatment.

3.4 | Paired samples model

The paired samples model included 754 admissions of 377 patients who were treated with and without ECT during two separate admissions. Characteristics of this population are presented in Table 4. ECT was used at the first admission in 124 (32.9%) of cases. Within 30 days after admission, 71 (19.0%) patients treated with ECT and 90 (24.1%) patients without ECT were readmitted (Table 5). Hazard ratio of readmission within 1 month for admissions with ECT was 0.75 (95% CI 0.55–1.03, $p = 0.075$) in univariate Cox regression analysis and 0.75 (95% CI 0.55–1.02, $p = 0.067$) in the multivariable Cox model compared to admissions without ECT (Table 6). Results of Cox regression analyses of associations between ECT and time to readmission for different subgroups are presented in Table 7. None of the subgroups differed in outcomes after ECT treatment. A Kaplan–Meier graph describing readmissions during the first year after index admissions with and without ECT is presented in Figure 2. Comparison of readmission within 1 year after hospitalizations with ECT and without ECT using the Log Rank (Mantel-Cox) test showed no statistical significance ($p = 0.085$).

3.5 | Propensity score matching analysis

The associations between admissions with ECT ($n = 902$) and time to readmission, as compared to admissions

TABLE 5 Readmission in different periods in individuals treated with or without ECT during separate mania episodes.

	ECT treated	Without ECT
Readmission within 1 month	19.0% (71 of 374)	24.1% (90 of 374)
Readmission within 3 months	33.7% (122 of 362)	38.9% (145 of 373)
Readmission within 6 months	44.5% (159 of 357)	48.5% (179 of 369)
Readmission within 12 months	59.8% (208 of 348)	63.7% (233 of 364)

Abbreviation: ECT, electroconvulsive therapy.

TABLE 6 Results of the Cox regression analysis of risk factors for rehospitalization or suicide within 30 days after admission for mania for patients treated with ECT and without ECT during two different admissions.

		Univariate		Multivariable	
		HR (95% CI)	<i>p</i> -value	aHR (95% CI)	<i>p</i> -value
ECT during the index admission	No	Reference			
	Yes	0.75 (0.55–1.03)	0.075	0.75 (0.55–1.02)	0.067

Abbreviations: aHR, adjusted hazard ratio; CI, confidence interval; ECT, electroconvulsive therapy; HR, hazard ratio; SD, standard deviation.

without ECT ($n = 1804$) were analyzed. This analysis was in line with findings in the whole study population and did not reveal any significant associations (aHR 0.91, 95% CI 0.76–1.08, $p = 0.277$).

4 | DISCUSSION

In the main analysis of this study, ECT for a manic episode was not associated with a differential time to readmission compared with other antimanic treatment than ECT. However, in a model with paired samples, the risk of readmission during the first 30 days after discharge after admissions with ECT was 0.75 (95% CI 0.55–1.02) compared with admissions without ECT, showing a trend toward a better outcome after ECT ($p = 0.067$). After admissions with ECT, 19% of patients were readmitted during the first month while 24.1% of admissions without ECT resulted in readmission. The results of this study suggest that a potential difference between time to readmission after hospitalizations with and without ECT may continue for a prolonged period after the ECT, as presented on the Kaplan–Meier curve (Figure 1) although the difference did not reach the statistical significance, with a Log Rank test p -value of 0.085.

There were some considerable differences between admissions with ECT and without ECT, as shown in Table 1. We hypothesize that differences between admissions with and without ECT in this model may result in a potential protective effect of ECT being underestimated. Patients treated with ECT had more previous admissions. A greater number of previous admissions has in multiple studies been associated with a higher risk of relapse.¹¹ Also in this study, the number of admissions before the index admission had the strongest association with time to readmission (Appendix C). Admissions with ECT were longer. The median time from the first admission day to the first ECT was 7 days, which suggests that patients treated with ECT had failed to show signs of response to the first-line treatment. We hypothesize that an unsatisfactory response to the first-line treatment might be a proxy for more severe illness. However, it is difficult to demonstrate causality. Results from different studies that

TABLE 7 Results of multivariable Cox regression analysis of association between ECT treatment and time to rehospitalization or suicide within 30 days for different subpopulations in paired sample.

	aHR	P-value	Interaction
Sex			
Male	0.74 (0.46–1.20)	0.224	Reference
Female	0.73 (0.48–1.11)	0.141	0.958
Age			
Age 14–45	0.80 (0.51–1.27)	0.347	Reference
Age 46–87	0.68 (0.44–1.11)	0.092	0.603
Education			
Less than high school	0.33 (0.14–0.80)	0.014	Reference
High school	0.86 (0.56–1.33)	0.498	0.302
Some college <3 yrs	2.23 (0.58–8.52)	0.242	0.605
College ≥ 3 yrs	0.70 (0.33–1.46)	0.340	0.277
Cohabiting			
Cohabiting 0	0.73 (0.48–1.11)	0.145	Reference
Cohabiting 1	0.75 (0.46–1.23)	0.255	0.719
Psychosis			
No	0.61 (0.36–1.01)	0.056	Reference
Yes	0.82 (0.54–1.25)	0.360	0.502
Substance use disorder			
No	0.75 (0.53–1.07)	0.114	Reference
Yes	0.50 (0.22–1.13)	0.097	0.702
Anxiety			
No	0.69 (0.47–1.02)	0.060	Reference
Yes	0.98 (0.54–1.77)	0.946	0.497
Personality disorder			
No	0.75 (0.54–1.04)	0.088	Reference
Yes	0.56 (0.18–1.71)	0.308	0.908
OCD			
No	0.75 (0.54–1.03)	0.073	Reference
Yes	10.85 (0.39–304.19)	0.161	0.897
ADHD			
No	0.76 (0.55–1.04)	0.086	Reference
Yes	815.65 (1.46–456,989)	0.038	0.649
Autism			
No	0.72 (0.52–0.99)	0.043	Reference
Yes	>250 × 10 ⁶	0.409	0.215
Number admissions			
0–5	0.50 (0.25–1.00)	0.051	Reference
6–20	1.25 (0.60–2.59)	0.558	0.479
>20	0.74 (0.48–1.14)	0.173	0.265
Lithium			
No	0.54 (0.32–0.94)	0.028	Reference
Yes	0.88 (0.59–1.32)	0.543	0.250

(Continues)

TABLE 7 (Continued)

	aHR	P-value	Interaction
Valproate			
No	0.64 (0.43–0.94)	0.024	Reference
Yes	1.09 (0.61–1.92)	0.777	0.210
Lamotrigine			
No	0.76 (0.54–1.08)	0.124	Reference
Yes	0.57 (0.22–1.48)	0.246	0.767
Benzodiazepines			
No	0.68 (0.47–0.98)	0.037	Reference
Yes	1.31 (0.66–2.63)	0.443	0.211
LAAs			
No	0.77 (0.55–1.07)	0.119	Reference
Yes	0.38 (0.11–1.24)	0.108	0.611
Oral antipsychotics			
No	0.79 (0.42–1.48)	0.463	Reference
Yes	0.71 (0.49–1.03)	0.074	0.631
Antidepressants			
No	0.79 (0.42–1.48)	0.463	Reference
Yes	0.71 (0.49–1.03)	0.074	0.745

Abbreviations: ADHD, attention deficit hyperactivity disorder; aHR, adjusted hazard ratio; LAAs, long-acting antipsychotics; OCD, obsessive compulsive disorder; SD, standard deviation.

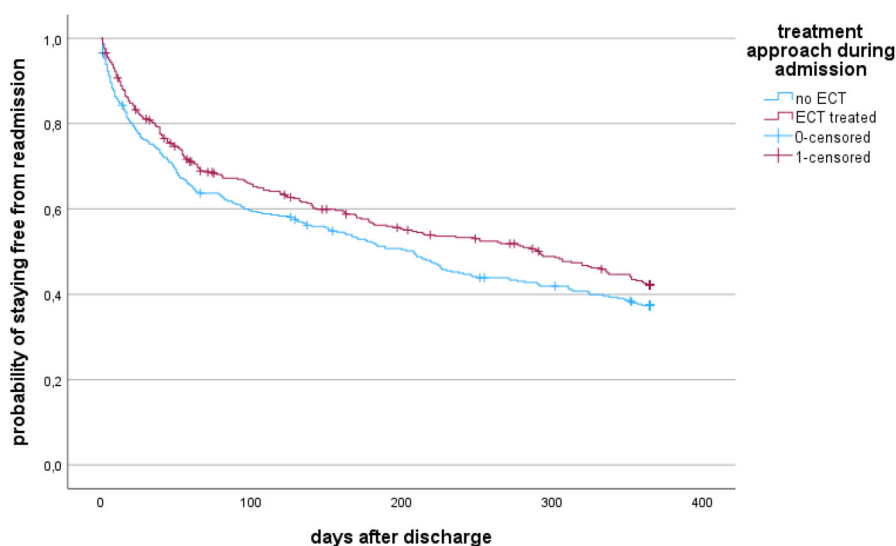


FIGURE 2 Kaplan-Meier survival analysis of admissions with and without ECT.

investigate the association between the length of psychiatric admissions and the rate of rehospitalization are inconsistent.^{22–26} Some indicate higher readmission rates in patients with shorter hospital stays^{22–26} while others find no such association.^{23,24}

In this study, death by suicide occurred after 26 out of 12,337 admissions, and only one of these was an admission with ECT. Some studies suggest that ECT has an

anti-suicidal effect in patients treated for severe depression.²⁷ However, the results of the present study should be interpreted with caution due to the low event rate and the fact that patients were treated without ECT in 92.7% of admissions.

The majority of patients who were treated with ECT had unilateral electrode placement and the median number of treatment sessions was 6. Despite that most

patients were assessed as very much or much improved after ECT (86%), some patients were possibly treated with suboptimal ECT.

In this study based on a Swedish population, ECT was used to treat mania in 7.3% of cases. By comparison, a Swedish study that investigated the association between ECT and readmission in patients with moderate to severe major depressive episodes showed that ECT was used in 26.8% of admissions.¹⁵ On the other hand, the study by Slade et al.⁷ that investigated psychiatric readmission of patients with MDD, bipolar disorder, or schizoaffective disorder in 9 states in the US showed that ECT was used in 1.5% of inpatients. These numbers reflect a wide variability in the usage of ECT across countries and diagnoses. Many authors have discussed the multifactorial background of these differences.^{28,29} Given the efficacy of ECT, this treatment method is probably underused, at least in some parts of the world. There is a need for more studies, especially randomized controlled trials to optimize the use of ECT in bipolar disorder.

5 | LIMITATIONS

The results of this study may be affected by the indication bias, as there was no randomization in this study. No mania specific rating scales were used to assess the efficacy of the treatments used during index admissions. Information about psychopharmacological treatment did not contain information about dosages or compliance. No information about ECT continuation/maintenance treatment was available. Additional continuation/maintenance ECT might have contributed to even longer time to readmission.

6 | CONCLUSION

Time to readmission within 30 days after discharge in patients with mania who were treated with ECT did not differ significantly from patients treated without ECT. However, in a paired sampled model, there was a trend toward longer time to readmission in ECT treated patients. Patients treated with ECT had more severe lifetime illness.

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CONFLICT OF INTEREST STATEMENT

Mikael Landén has received lecture honoraria for Lundbeck pharmaceuticals and served as consultant for Astra-Zeneca. The other authors report no conflict of interest.

PEER REVIEW

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1111/acps.13689>.

DATA AVAILABILITY STATEMENT

Research data are not shared.

ORCID

Katarzyna Popiolek  <https://orcid.org/0000-0002-6883-9377>

Axel Nordenskjöld  <https://orcid.org/0000-0001-7454-3065>

REFERENCES

1. Excellence NifHaC. Bipolar disorder (update): the management of bipolar disorder in adults, children and adolescents in primary and secondary care. www.nice.org.uk/guidance/CG185/InformationForPublic 2014.
2. Goodwin GM, Haddad PM, Ferrier IN, et al. Evidence-based guidelines for treating bipolar disorder: revised third edition recommendations from the british association for psychopharmacology. *J Psychopharmacol*. 2016;30:495-553.
3. Grunze H, Vieta E, Goodwin GM, et al. The world Federation of Societies of biological psychiatry (WFSBP) guidelines for the biological treatment of bipolar disorders: update 2009 on the treatment of acute mania. *World J Biol Psychiatry*. 2009;10:85-116.
4. Yatham LN, Kennedy SH, Parikh SV, et al. Canadian network for mood and anxiety treatments (CANMAT) and International Society for Bipolar Disorders (ISBD) collaborative update of CANMAT guidelines for the management of patients with bipolar disorder: update 2013. *Bipolar Disord*. 2013;15:1-44.
5. Popiolek K, Bejerot S, Landen M, Nordenskjöld A. Association of Clinical and Demographic Characteristics with Response to electroconvulsive therapy in mania. *JAMA Netw Open*. 2022;5:e2218330.
6. Mukherjee S, Sackeim HA, Schnur DB. Electroconvulsive therapy of acute manic episodes: a review of 50 years' experience. *Am J Psychiatry*. 1994;151:169-176.
7. Slade EP, Jahn DR, Regenold WT, Case BG. Association of Electroconvulsive Therapy with Psychiatric Readmissions in US hospitals. *JAMA Psychiatry*. 2017;74:798-804.
8. Hong J, Reed C, Novick D, Haro JM, Windmeijer F, Knapp M. The cost of relapse for patients with a manic/mixed episode of bipolar disorder in the EMBLEM study. *Pharmacoeconomics*. 2010;28:555-566.

9. Perlis RH, Ostacher MJ, Patel JK, et al. Predictors of recurrence in bipolar disorder: primary outcomes from the systematic treatment enhancement program for bipolar disorder (STEP-BD). *Am J Psychiatry*. 2006;163:217-224.
10. Gignac A, McGirr A, Lam RW, Yatham LN. Recovery and recurrence following a first episode of mania: a systematic review and meta-analysis of prospectively characterized cohorts. *J Clin Psychiatry*. 2015;76:1241-1248.
11. Li DJ, Lin CH, Wu HC. Factors predicting re-hospitalization for inpatients with bipolar mania—a naturalistic cohort. *Psychiatry Res*. 2018;270:749-754.
12. Woo YS, Bahk WM, Jung YE, et al. One-year rehospitalization rates of patients with first-episode bipolar mania receiving lithium or valproate and adjunctive atypical antipsychotics. *Psychiatry Clin Neurosci*. 2014;68:418-424.
13. Ahmadi N, Moss L, Simon E, Nemeroff CB, Atre-Vaidya N. Efficacy and long-term clinical outcome of comorbid posttraumatic stress disorder and major depressive disorder after electroconvulsive therapy. *Depress Anxiety*. 2016;33:640-647.
14. Tor PC, Bin AE. Mirror readmission study of the Association of Electroconvulsive Therapy with 1-year mood disorder readmissions in a tertiary mood disorder unit. *J ECT*. 2020;36:111-114.
15. Stenmark L, Kellner CH, Landen M, Larsson I, Møghina M, Nordenskjöld A. Electroconvulsive therapy and psychiatric readmission in major depressive disorder - a population-based register study. *Acta Psychiatr Scand*. 2021;144:599-625.
16. von Elm E, Altman DG, Egger M, et al. The strengthening of reporting of observational studies in epidemiology (STROBE) statement: guidelines for reporting observational studies. *Lancet*. 2007;370:1453-1457.
17. Organisation WH. *International Statistical Classification of Diseases and Related Health Problems, 10th Revision (ICD-10)*. 5th ed. World Health Organisation; 2016.
18. Ahmad I, Sandberg M, Brus O, et al. Validity of diagnoses, treatment dates, and rating scales in the Swedish national quality register for electroconvulsive therapy. *Nord J Psychiatry*. 2022;76:96-103.
19. Ludvigsson JF, Andersson E, Ekbom A, et al. External review and validation of the Swedish national inpatient register. *BMC Public Health*. 2011;11:450.
20. Ludvigsson JF, Svedberg P, Olen O, Bruze G, Neovius M. The longitudinal integrated database for health insurance and labour market studies (LISA) and its use in medical research. *Eur J Epidemiol*. 2019;34:423-437.
21. Wallerstedt SM, Wettermark B, Hoffmann M. The first decade with the Swedish prescribed drug register—a systematic review of the output in the scientific literature. *Basic Clin Pharmacol Toxicol*. 2016;119:464-469.
22. Heeren O, Dixon L, Gavirneni S, Regenold WT. The association between decreasing length of stay and readmission rate on a psychogeriatric unit. *Psychiatr Serv*. 2002;53:76-79.
23. Thomas MR, Rosenberg SA, Giese AA, Fryer GE, Dubovsky SL, Shore JH. Shortening length of stay without increasing recidivism on a university-affiliated inpatient unit. *Psychiatr Serv*. 1996;47:996-998.
24. Edward-Chandran T, Malcolm DE, Bowen RC. Reduction of length of stay in an acute care psychiatric unit. *Can J Psychiatry*. 1996;41:49-51.
25. Appleby L, Desai PN, Luchins DJ, Gibbons RD, Hedeker DR. Length of stay and recidivism in schizophrenia: a study of public psychiatric hospital patients. *Am J Psychiatry*. 1993;150:72-76.
26. Lieberman PB, Wiitala SA, Elliott B, McCormick S, Goyette SB. Decreasing length of stay: are there effects on outcomes of psychiatric hospitalization? *Am J Psychiatry*. 1998;155:905-909.
27. Ronnqvist I, Nilsson FK, Nordenskjöld A. Electroconvulsive therapy and the risk of suicide in hospitalized patients with major depressive disorder. *JAMA Netw Open*. 2021;4:e2116589.
28. Rakita U, Bingham K, Fung K, Giacobbe P. Factors associated with global variability in electroconvulsive therapy utilization. *J ECT*. 2017;33:253-259.
29. Leiknes KA, Jarosh-von Schweder L, Hoie B. Contemporary use and practice of electroconvulsive therapy worldwide. *Brain Behav*. 2012;2:283-344.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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